a)

KH*

Null hypothesis H_0: $E[\delta_S] = 0$
Alternative hypothesis H_A: $E[\delta_S] \neq 0$

Input Gene tree set $G = \{g_1, g_2, g_3, \ldots, g_l\}$
Choose Significance level α

Algorithm 1: KH* with non-parametric bootstrapping (KH_1*)
gene tree analogy: *priNPfcd*

Compute test statistic $\delta = \ln L(S_1) - \ln L(S_2) = L_1 - L_2$
for $i = 1$ to b:
  Resample with replacement to get $G^{(i)}$ (bootstrapped sample of G)
  Optimize MSC parameters to maximize the log-likelihoods $L_1{}^{(i)}$ and $L_2{}^{(i)}$ for species topologies S_1 and S_2, respectively
  Compute bootstrapped statistic $\delta^{(i)} = L_1{}^{(i)} - L_2{}^{(i)}$
Center the distribution of $\delta^{(i)}$s by subtracting the mean $\delta^{(i)}$ from each $\delta^{(i)}$
Return p-value given the null distribution of $\delta^{(i)}$
Reject null hypothesis **if** p-value $\leq \alpha$

Algorithm 2: KH* with RELL bootstrapping (KH_2*)
gene tree analogy: *priNPncd*

Optimize MSC parameters independently for S_1 and S_2
Compute test statistic $\delta = \ln L(S_1) - \ln L(S_2) = L_1 - L_2$
for $i = 1$ to b:
  Resample with replacement to get $G^{(i)}$
  Calculate log-likelihoods $L_1{}^{(i)}$ and $L_2{}^{(i)}$ using optimized parameters from Step 1 (RELL approach)
  Compute bootstrapped statistic $\delta^{(i)} = L_1{}^{(i)} - L_2{}^{(i)}$
Center the distribution of $\delta^{(i)}$s by subtracting the mean $\delta^{(i)}$ from each $\delta^{(i)}$
Return p-value given the null distribution of $\delta^{(i)}$s
Reject null hypothesis **if** p-value $\leq \alpha$

Algorithm 3: KH* with RELL and Normal approximation (KH_3*)
gene tree analogy: *priNPncn*

Optimize MSC parameters independently for S_1 and S_2
Compute test statistic $\delta = \ln L(S_1) - \ln L(S_2) = L_1 - L_2$
for $i = 1$ to b:
  Resample with replacement to get $G^{(i)}$
  Calculate log-likelihoods $L_1{}^{(i)}$ and $L_2{}^{(i)}$ using optimized parameters from Step 1 (RELL approach)
  Compute bootstrapped statistic $\delta^{(i)} = L_1{}^{(i)} - L_2{}^{(i)}$
Center the distribution of $\delta^{(i)}$s by subtracting the mean $\delta^{(i)}$ from each $\delta^{(i)}$
Calculate the variance v^2 of the $\delta^{(i)}$ distribution
Return p-value assuming a $N(0, v^2)$
Reject null hypothesis **if** p-value $\leq \alpha$

b)

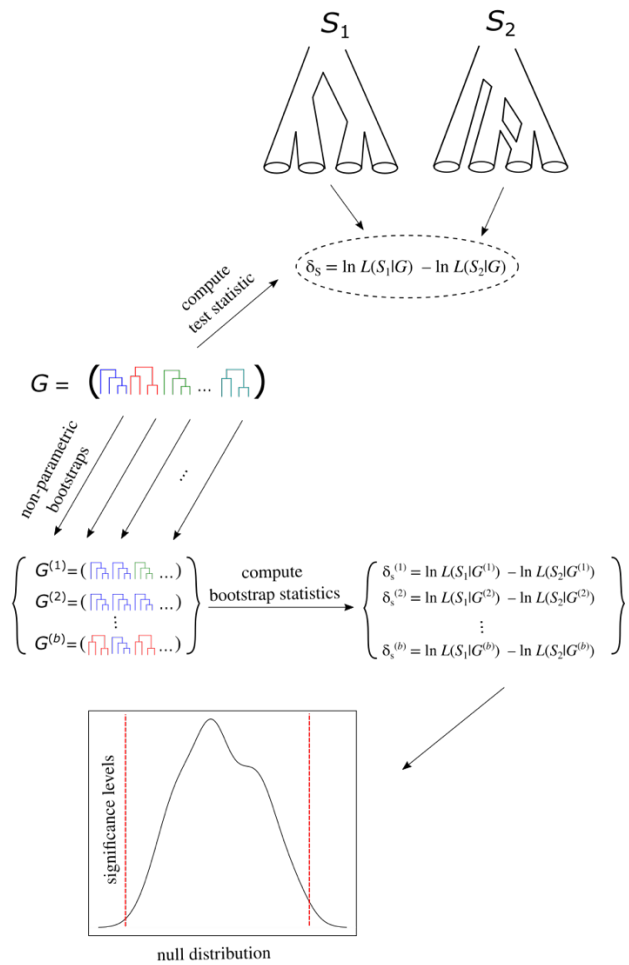

FIGURE S1. The KH* test for species-level hypotheses. Details for the three algorithms (KH_1^*, KH_2^*, and KH_3^*) are provided in panel (a) and a general schematic overview is shown in panel (b). Briefly, the KH* test evaluates whether the difference in MSC likelihoods δ_S computed between two species topologies S_1 and S_2 (top right) is a plausible draw from a null distribution (bottom right) obtained using non-parametric bootstrapping (center right). Two example topologies are shown in panel (b): a classical bifurcating topology on the left (S_1) and a species network on the right (S_2). Non-parametric bootstrapping of the input gene tree set G is conducted to obtain b total replicates $G^{(1)}$, $G^{(2)}$, …, $G^{(b)}$, which, in turn, yields a distribution of $\delta_S^{(1)}$, $\delta_S^{(2)}$, …, $\delta_S^{(b)}$ under the null hypothesis. The primary difference between the three KH* algorithms is whether RELL bootstrapping is used (KH_2^*, and KH_3^*) or not (KH_1^*), while the KH_3^* also uses normal approximation to evaluate significance.

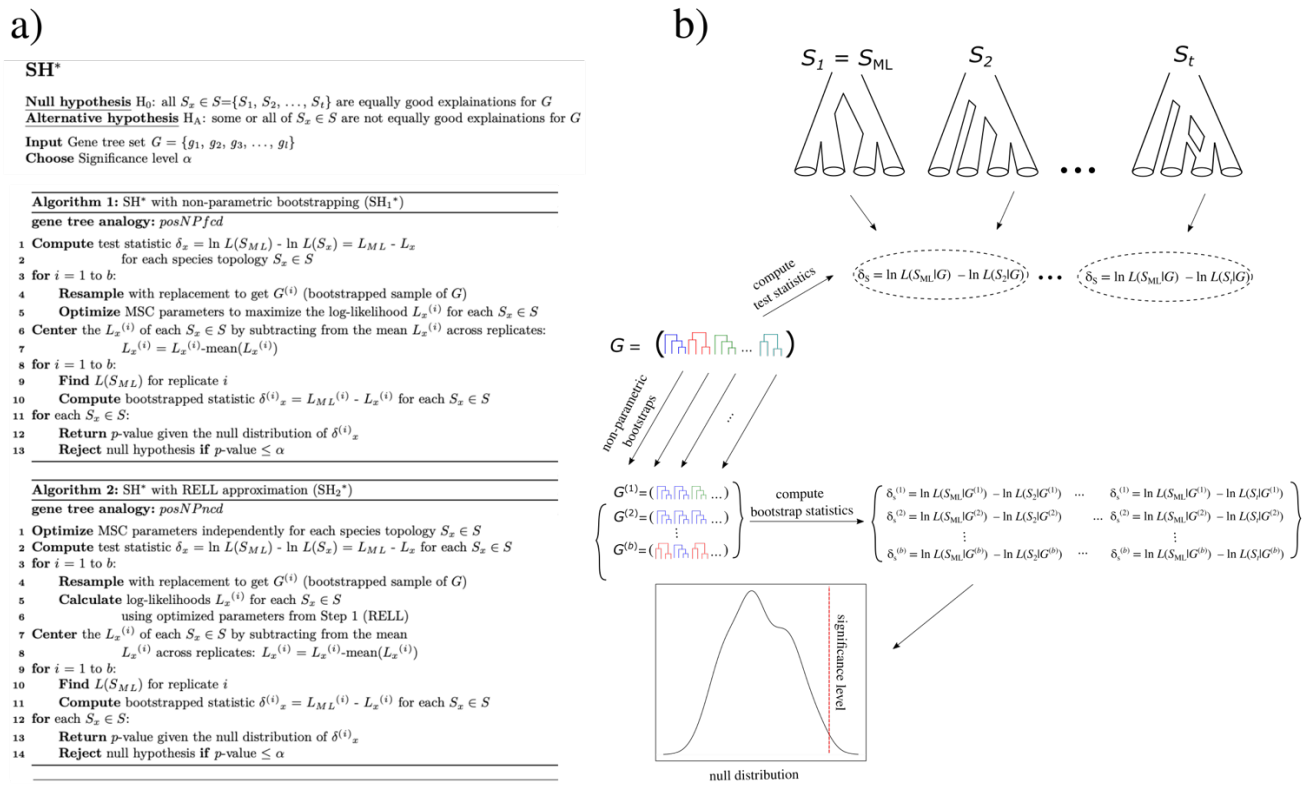

FIGURE S2. The SH* test for species-level hypotheses. Details for the two algorithms (SH_1^*, and SH_2^*) are provided in panel (a) and a general schematic overview is shown in panel (b). Briefly, the SH* test evaluates whether the difference in MSC likelihoods δ_S computed between two or more species topologies $S_1, S_2, \ldots, S_t$ (top right) included in a set of t topologies is a plausible draw from a null distribution (bottom right) obtained using non-parametric bootstrapping (center right). Specifically, the difference in MSC likelihood is computed between the species topology in the set with maximum likelihood (S_{ML}) and each of the other $t - 1$ topologies. Several example topologies are shown in panel (b), which include two classical bifurcating trees on the left ($S_1 = S_{\mathrm{ML}}$ and S_2) and a species network on the right (S_t). Non-parametric bootstrapping of the input gene tree set G is conducted to obtain b total replicates $G^{(1)}$, $G^{(2)}$, …, $G^{(b)}$, which, in turn, yields a distribution of $\delta_S^{(1)}$, $\delta_S^{(2)}$, …, $\delta_S^{(b)}$ under the null hypothesis. Each round of non-parametric bootstrapping is followed by pairwise likelihood difference computations between each topology and the maximum likelihood topology in the set to obtain $\delta_S^{(i)}$. The set of t topologies includes the maximum likelihood estimate $S_1 = S_{\mathrm{ML}}$, which is used to contrast against the alternative topologies. As with KH*, the two SH* algorithms differ on whether RELL bootstrapping is used (SH_2^*) or not (SH_1^*).

a)

SOWH*

Null hypothesis H_0: S_1 is the true species topology
Alternative hypothesis H_A: some other topology is the true species topology

Input Gene tree set $G = \{g_1, g_2, g_3, \ldots, g_l\}$
Choose Significance level α

Algorithm 1: SOWH* with parametric bootstrapping (SOWH$_1$*)
gene tree analogy: *posPfud*

Optimize MSC parameters for S_1 and find the maximum likelihood topology S_{ML}
Compute test statistic $\delta = \ln L(S_{ML})$ - $\ln L(S_1)$
for $i = 1$ to b:
  Simulate a set of gene trees $G^{(i)}$ using the optimized MSC parameters for S_1 from Step 1
  Optimize MSC parameters for S_1 to get $L_1{}^{(i)}$
  Find the maximum likelihood topology S_{ML} to get $L_{ML}{}^{(i)}$
  Compute bootstrapped statistic $\delta^{(i)} = L_{ML}{}^{(i)}$ - $L_1{}^{(i)}$
Return p-value given the null distribution of $\delta^{(i)}$
Reject null hypothesis **if** p-value $\leq \alpha$

Algorithm 2: SOWH* with parametric bootstrapping and RELL (SOWH$_2$*)
gene tree analogy: *posPpud*

Optimize MSC parameters for S_1 and find the maximum likelihood topology S_{ML}
Compute test statistic $\delta = \ln L(S_{ML})$ - $\ln L(S_1)$
for $i = 1$ to b:
  Simulate a set of gene trees $G^{(i)}$ using the optimized MSC parameters for S_1 from Step 1
  Compute $L_1{}^{(i)}$ using the optimized MSC parameters for S_1 from Step 1
  Find the maximum likelihood topology S_{ML} to get $L_{ML}{}^{(i)}$
  Compute bootstrapped statistic $\delta^{(i)} = L_{ML}{}^{(i)}$ - $L_1{}^{(i)}$
Return p-value given the null distribution of $\delta^{(i)}$
Reject null hypothesis **if** p-value $\leq \alpha$

b)

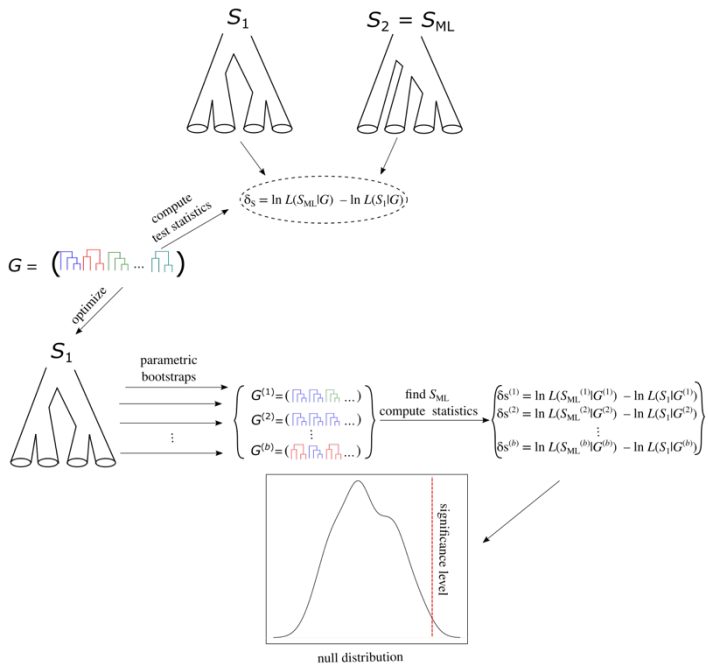

FIGURE S3. The SOWH* test for species-level hypotheses. Details for the two algorithms (SOWH_1^*, and SOWH_2^*) are provided in panel (a) and a general schematic overview is shown in panel (b). Briefly, the SOWH* test evaluates whether the difference in MSC likelihoods δ_S computed between a hypothesized focal species topology (S_1) and the maximum likelihood estimate (S_{ML}) is a plausible draw from a null distribution (bottom right) obtained using parametric bootstrapping (center right). Parametric bootstrapping is conducted using the optimized branch lengths of S_1 to obtain b total replicates $G^{(1)}$, $G^{(2)}$, …, $G^{(b)}$ that are used to find a maximum likelihood topology for each replicate which, in turn, yields a distribution of $\delta_S^{(1)}$, $\delta_S^{(2)}$, …, $\delta_S^{(b)}$ under the null hypothesis. Each round of parametric bootstrapping is followed by a search for a maximum likelihood topology that is used to compare with the target topology S_1 at their optimized parameter values (branch lengths) in order to compute the values of $\delta_S^{(i)}$. As with KH* and SH*, the two SH* algorithms differ on whether RELL bootstrapping is used (SOWH_2^*) or not (SOWH_1^*) in the processes of generating the null distribution of δ_S.